

UNIT IV
ASSIGNMENT

Class #5:

Introduction to Eigen values and Eigenvectors,

1) Find the eigenvalues and the corresponding eigenvectors of

$$(i) \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

$$(iii) \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

Answers:

$$(i) \quad 1, -2, -2 \quad k_1(1, -1, 1), k_2(-1, 1, 0), k_3(-1, 0, 1)$$

$$(ii) \quad 1, -2, -2 \quad k_1(1, -1, 1), k_2(-1, 1, 0)$$

$$(iii) \quad 0, 3, 15 \quad k_1(1, 2, 2), k_2(2, 1, -2), k_3(2, -2, 1)$$